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Federated Traffic Intelligence: AI-powered traffic prediction through collaborative learning

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Problem Description

Problem Statement: Federated Traffic-Flow Prediction

- Traffic data is distributed across different owners, such as city agencies, highway operators, and private companies.
- Centralizing raw traffic data can be costly, time-consuming, and restricted.
- Traffic prediction requires analysing historical data to forecast future traffic flow.
- Limited network bandwidth makes transferring large datasets inefficient.
- Different locations may have different traffic patterns, making collaborative model training challenging.
- A solution is needed to balance prediction accuracy, communication cost, and model convergence without pooling raw data.

Background

Background and Proposed Optimization

Traffic data is distributed across multiple owners, including city agencies, highway operators, and private companies. Centralizing this data can be costly, time-consuming, and restricted.

- **Data Distribution:** Traffic records are stored at different locations.
- **Communication Cost:** Transferring large datasets requires significant network resources.
- **Traffic Prediction:** Historical traffic patterns can help forecast future traffic flow.
- **Different Traffic Patterns:** Each location may have unique data distributions.
- **Proposed Optimization:** Federated Learning enables clients to train locally and share model updates instead of raw data.
- **Efficient Collaboration:** A GRU model with Federated Averaging (FedAvg) will be evaluated for prediction accuracy, communication cost, and convergence.

Literature & RFC Review

Related Work and Existing Approaches

- **Centralized Traffic Prediction:** Requires collecting data centrally, increasing communication costs and data-sharing concerns.
- **Federated Learning (McMahan et al., 2017):** Uses FedAvg to train a shared model without exchanging raw data.
- **GRU Networks (Cho et al., 2014):** Learn sequential patterns for time-series prediction.
- **Communication-Efficient Learning:** Focuses on reducing communication overhead during model training.

Research Gap

Our project combines GRU and FedAvg to compare centralized and federated traffic prediction under IID and non-IID data, evaluating **MAE, RMSE, communication cost, and convergence**.

References

- McMahan et al. (2017). *Communication-Efficient Learning of Deep Networks from Decentralized Data.*
- Cho et al. (2014). *Learning Phrase Representations using RNN Encoder–Decoder.*
- Li et al. (2020). *Federated Optimization in Heterogeneous Networks.*

Proposed Solution

Our solution combines **GRU**-based traffic prediction with Federated Averaging **(FedAvg)** to enable collaborative learning without sharing raw traffic data.

- **Data Preprocessing:** Clean and prepare the public PeMS04 traffic dataset.
- **Client Simulation:** Divide data among four simulated clients.
- **Local Training:** Each client uses a GRU model to predict traffic flow 15 minutes ahead using the previous hour of data.
- **Federated Learning:** The server aggregates client model updates using FedAvg and redistributes the global model over multiple rounds.
- **Evaluation:** Compare centralized and federated learning using MAE, RMSE, communication cost, and convergence.
- **Dashboard:** Display experimental comparisons using Streamlit.
- **Expected Outcome:** A working prototype evaluating traffic prediction accuracy and communication efficiency under different data distributions.

Optimization Proposed

Aspect	Before Optimization	Proposed Enhancement
Data Sharing	Raw data centralized on a server	Keep raw data local using federated learning
Model Training	Training on pooled data	Local training across four simulated clients
Traffic Prediction	General forecasting model	GRU model for time-series prediction
Model Aggregation	No collaborative aggregation	FedAvg combines client model updates
Communication	Transfer large datasets	Exchange compact model updates
Data Distribution	Assumes similar traffic patterns	Test IID and non-IID client data
Performance Evaluation	Focus on prediction accuracy	Measure MAE, RMSE, communication cost, and convergence
Result Visualization	Limited comparison of results	Streamlit dashboard for performance analysis

Solution Architecture & Design

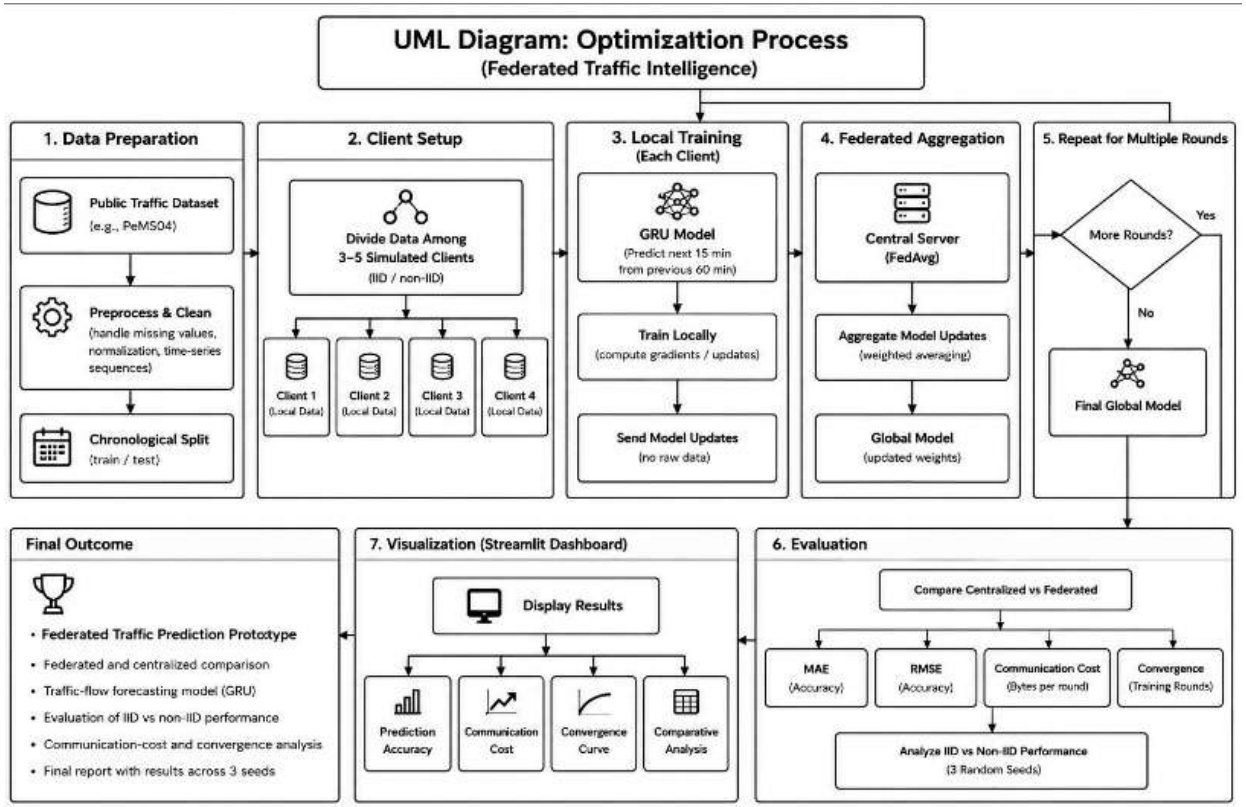


Figure: System Architecture UML Diagram

Data Flow Diagram

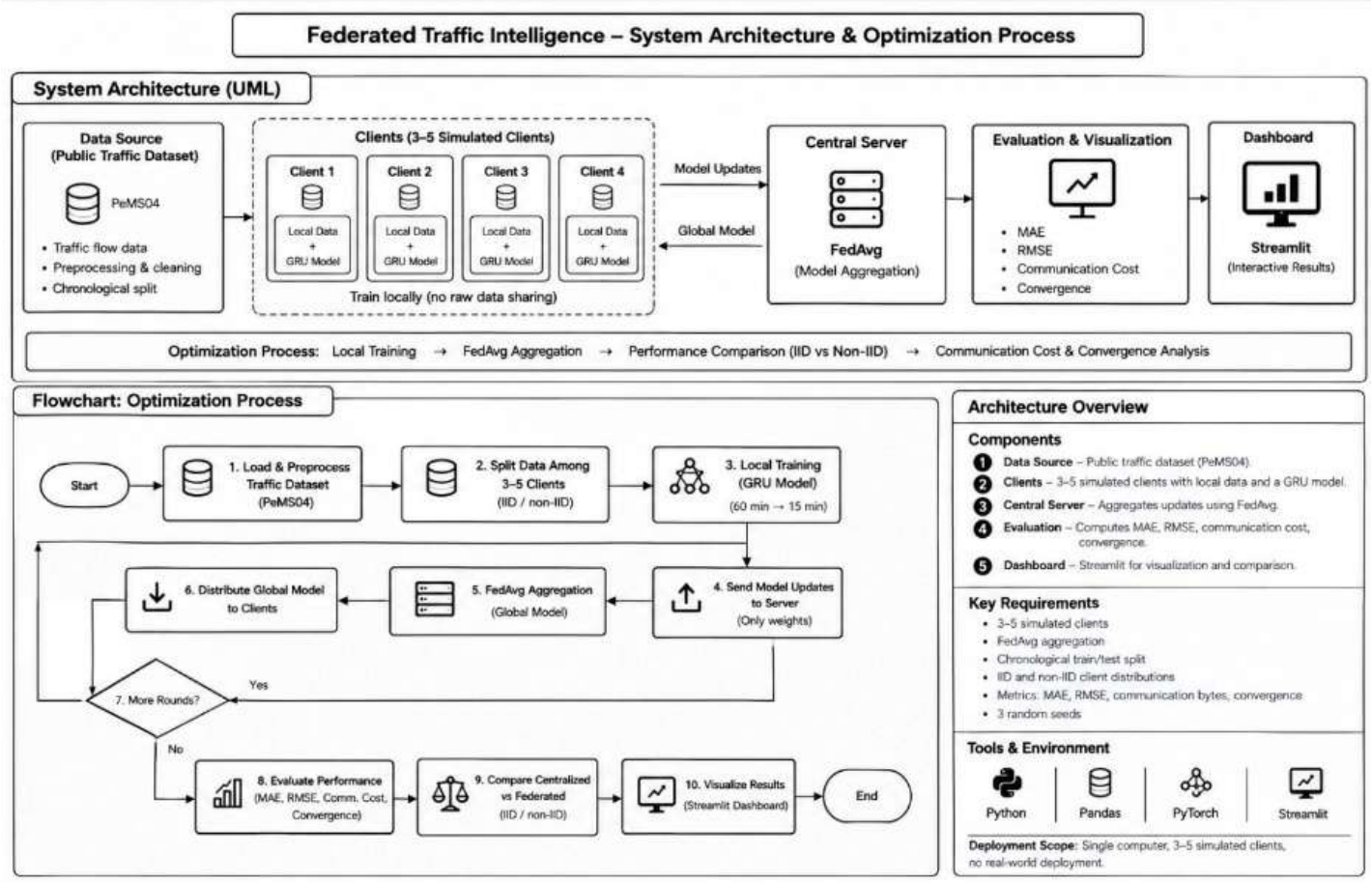


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Deployment Methodology

- **Data Preparation:** Clean and preprocess the Vellore traffic dataset.
- **Client Setup:** Divide data among four simulated clients using IID and non-IID distributions.
- **Model Training:** Train local GRU models to predict the next 15 minutes using the previous 60 minutes.
- **Federated Aggregation:** Apply FedAvg to combine model updates without sharing raw data.
- **Evaluation & Visualization:** Compare centralized and federated models using MAE, RMSE, communication cost, and convergence; display results using Streamlit.

Deployment Milestones

Phase	Key Deliverable
1. Data Preparation	Cleaned Vellore traffic dataset
2. Model Development	Working GRU baseline
3. FL Implementation	Four clients + FedAvg
4. Testing & Evaluation	Metrics across IID/non-IID scenarios
5. Dashboard & Validation	Streamlit dashboard and final report

Tools: Python · Pandas · PyTorch · Streamlit
Deployment Scope: Single computer, four simulated clients, three experimental runs; no real-world deployment.

Bibliography I

1. McMahan, B. et al. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics* (2017), pp. 1273–1282.
2. Li, T. et al. “Federated Optimization in Heterogeneous Networks.” In: *Proceedings of Machine Learning and Systems* 2 (2020), pp. 429–450.
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Thank You for Your Attention!